

Parent Guide — middle / module-09- subtraction-nikhilam

IKS Curriculum

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Parent Guide — Module 9: Subtraction by Complements (*Nikhilam Sūtra*)

Your child's age: Middle (Grades 6–8) · **Module length:** 10 sessions × 45 min

What your child is learning

A mental-arithmetic technique called the **Nikhilam method** — a way of subtracting from round numbers (10, 100, 1000, 10 000, ...) without any borrowing.

The rule, in Sanskrit, is six words:

Nikhilam Navataścaramam Daśataḥ — “all from nine and the last from ten.”

In practice: to find $1000 - 567$, your child looks at each digit of 567 and reads off:

- $9 - 5 = 4$
- $9 - 6 = 3$
- $10 - 7 = 3$

Answer: **433**. No borrowing. They check it: $567 + 433 = 1000$. ✓

This generalises to any power of ten, and (with a small adjustment) to general subtractions like $8243 - 567$.

A note on “Vedic mathematics”

Important honest framing:

- The Nikhilam method is a real, useful technique. It works exactly as advertised.
- The sūtra was published in 1965 by Bharati Krishna Tirthaji (1884–1960), Śaṅkarācārya of Govardhana Maṭha (Puri). His book *Vedic Mathematics* gives 16 sūtras and credits them to a Vedic appendix-text.
- **No other Sanskritist has located that text.** The method is real; the direct Vedic attribution is unverified.
- We teach the method enthusiastically AND are honest about its history. At Senior band your child reads the scholarly debate (Dani 1993) and forms their own view.

What to expect this fortnight

Daily warm-up: 1 minute of “complement to 10” — instant recall of digit-partners that add to 10.

Day 1: Baseline timing. Your child solves $10\ 000 - 4567$ using their normal method. They write down the time. Day 5 retest is on the same problem with Nikhilam; the visible speedup is the motivation.

Days 2–4: Find the complement (*pūraka*) of any number to 100, 1000, 10 000. Apply it to subtractions like $1000 - 567$.

Day 6: Why the rule works. The algebraic statement: $10^n - A = (10^n - 1) - A + 1$, where $(10^n - 1)$ is “all nines” and “+1” is the last-from-ten step.

Day 7: A surprising connection — binary computers use the same idea to subtract. It’s called *two’s complement*. Your child sees that the Nikhilam principle is alive inside every smartphone.

Days 8–10: Project work and final assessment.

How to help

Helpful:

- 5-minute home drill: call out two-digit numbers; your child gives the complement to 100. ($73 \rightarrow 27$. $41 \rightarrow 59$. $28 \rightarrow 72$.) Then 3-digit complements to 1000.
- Be the “customer” in a shopkeeper role-play. You give a 1000-rupee note for a 367-rupee item; your child gives change of 633. They explain how they got it.
- Try the method yourself. Most adults find it eye-opening.

Not helpful:

- Insisting that “the way I was taught (borrowing)” is the only right way. Both methods give the same answer; Nikhilam is faster when subtracting from a round base.
- Doing the project for them. Even the chart or video has to be in their handwriting / their voice.

What success looks like

By end of module, your child can:

- Recite the Nikhilam sūtra in Sanskrit and translate it.
- Find the complement of any 3-digit number to 1000 in under 3 seconds.
- Subtract from a power of ten with no written borrowing.
- Explain *why* the method works (a one-paragraph justification using place value).
- Distinguish the method (Tirthaji’s, published 1965) from a vague claim of “Vedic antiquity.”

Project options

Your child chooses one of:

1. **A shopkeeper's quick-change chart** — a one-page reference card a real shopkeeper could use to give change from 100, 500, or 1000 rupee notes instantly.
2. **A 60-second teaching video** — explaining Nikhilam to a Grade 4 student.
3. **A two's-complement explainer poster** — connecting Nikhilam to how computers subtract in binary.

Citation standard

Projects must cite at least one source. The standard citation is:

Tirthaji, Vedic Mathematics (1965), sūtra Nikhilam.

The full citation chain is documented in the module's `sources.md`.

Want to go deeper?

The Senior band (Grades 9–12) generalises Nikhilam to any base and unpacks the historicity question in detail. Preview:

`curriculum/senior/module-09-subtraction-nikhilam/README.md`.

10-Day Lesson Plan

Module 9 (Middle) — 10-Day Lesson Plan

At a glance

Day	Session title	Lesson file	Activity / assessment
1	The Sūtra and the Hook	<code>days/day-01-hook.md</code>	Baseline timed test
2	Complement to 100	<code>days/day-02-complement-100.md</code>	Pair drills
3	Complement to 1000	<code>days/day-03-complement-1000.md</code>	Formative quiz Part A
4	Subtracting from a Power of Ten	<code>days/day-04-power-of-ten.md</code>	<code>activities/activity-01-complement-drill.md</code>
5	Mid-module Check	<code>days/day-05-mid-module-check.md</code>	Formative quiz Part B
6	Why It Works	<code>days/day-06-</code>	Reflection paragraph

Day	Session title	Lesson file	Activity / assessment
		why-it-works.md	
7	Two's Complement Connection	days/day-07-tvos-complement.md	Cross-disciplinary note
8	General Subtraction with Complements	days/day-08-general-subtraction.md	activities/activity-02-shopkeeper.md
9	Speed Practice + Comparison	days/day-09-speed-comparison.md	Comparison table
10	Final Assessment + Project Share	days/day-10-final-assessment.md	quizzes/summative.md + project

Pacing notes

- **Day 1** establishes a baseline: students time themselves on $10000 - 4567$ using standard borrowing. **Day 5** retests the same problem with the Nikhilam method. The visible speedup is the module's motivation.
- **Days 2–4** are the core skill: finding complements and applying them to power-of-ten subtraction. If the class is solid by Day 3, Day 4's drill can be extended.
- **Day 6** ("Why it works") and **Day 7** (two's complement) can be reordered — some classes find the binary parallel more motivating than the algebra; in that case do Day 7 first, then loop back to algebra on Day 8 morning.
- **Days 8–9** introduce the general case and revisit speed. Day 10 is summative.

Daily warm-up (recurring)

Every class starts with **1 minute of "complement to 10" fluency**: teacher calls out a digit; students call back the partner that makes 10. ($8 \rightarrow 2$, $6 \rightarrow 4$, $9 \rightarrow 1$, etc.) Same warm-up as Module 5 — it is the prerequisite reflex for Nikhilam.

What students leave the module with

- Instant recall of complements to 10, 100, and 1000 for any number under 1000.
- Confidence subtracting from a power of ten without writing a borrow.
- An algebraic statement (in their own words) of why the method works: $10^n - A = (10^n - 1) - A + 1$, and the digit-wise rule applies because $10^n - 1$ is "all nines."
- One real-world application — change-counting, deficit checks, or a binary parallel.
- One project artefact they can share with a younger student.

Homework

Homework — Module 9 Nikhila Subtraction (Middle)

Day 1 → Day 2

Task: Memorise the sūtra. Practice 10 complements to 100: $73 \rightarrow ?$, $41 \rightarrow ?$, $28 \rightarrow ?$, $86 \rightarrow ?$, $95 \rightarrow ?$, $17 \rightarrow ?$, $60 \rightarrow ?$, $52 \rightarrow ?$, $34 \rightarrow ?$, $89 \rightarrow ?$ Time goal: under 30 seconds total.

Time: 5 min.

Day 2 → Day 3

Task: Have a parent or sibling call out 20 two-digit numbers in 90 seconds; you say the complement to 100 for each. Track how many you got right.

Time: 5 min.

Day 3 → Day 4

Task: Find the complement to 1000 of each: 156, 423, 789, 802, 64 ($\rightarrow 064$), 7 ($\rightarrow 007$), 555, 900. Find the complement to 10 000 of each: 3 456, 1 027, 8 888, 75 ($\rightarrow 0075$), 999. Total time goal: under 2 minutes.

Time: 10 min.

Day 4 → Day 5

Task: 12 power-of-ten subtractions on the back of today's worksheet. Self-check by adding (number + your answer should equal the base).

Time: 15 min.

Day 5 → Day 6

Task: Submit your project choice + a 3-sentence plan. (One of: shopkeeper chart / teaching video / two's-complement poster.) Plus: 5 mixed subtractions — $1000 - 358$, $10\ 000 - 6\ 427$, $1000 - 72$, $100\ 000 - 12\ 345$, $10\ 000 - 9$.

Time: 15 min.

Day 6 → Day 7

Task: Find ONE moment in your daily life where you naturally “round to the nearest 100 / 1000” without thinking. Write 2 sentences. Example: rounding a restaurant tip, counting change, comparing prices.

Plus: redo today's algebra example with $10\ 000 - 4321$. Show the identity $10^n - A = (10^n - 1) - A + 1$ for this case.

Time: 10 min.

Day 7 → Day 8

Task: Convert $13 - 5$ to binary (4 bits) and solve via two's complement. Show your work.
Plus: one reflection sentence — what surprised you about the binary connection.

Time: 10 min.

Day 8 → Day 9

Task: 6 mixed general subtractions: $-7\,234 - 587$ (base 1000) - $9\,612 - 3\,458$ (base 10 000) - $5\,100 - 67$ (base 100) - $13\,405 - 2\,850$ (base 10 000) - $800 - 39$ (base 100) - $25\,000 - 4\,999$ (base 10 000)

For each, show your chosen base, the complement, and the answer.

Time: 20 min.

Day 9 → Day 10

Task: Final project touches. 5 minutes of sūtra recitation. Bring all project materials. Be ready to present in 3 minutes.

Time: 15 min plus material gathering.